PHYSLLM · G0 → N02 → N03

PhysLLM: Residual Dynamics at Model Scale

One measurement programme, three generations: a calibration gate, a frozen evaluation manifest, and MoE routing inside a 117B-parameter model — every number shipped with SHA-256 provenance.

> **Claim** — a residual stream can be measured the same way at every scale, and the protocol (gate → observables → manifest → gates → provenance) is what makes the numbers comparable across generations.

6
checkpoints calibrated (G0 gate)

14.4k
text x block rows captured

36
blocks · full / sliding attention

128/4
experts per token (top-4)

G0 calibration & lens equivalence › **N02** frozen observables + manifest › **N03** MoE routing-conditional

G0 · FOUNDATION

Calibration before claims

G0_Calibration_and_Lens_Equivalence v1.0.2, target **REAL-PASS**: prove the rig (hooks, dtype, eager/safe mode, tolerances) is **lens-equivalent** across architectures before any science claim.

CHECKPOINT	DTYPE · EAGER	TOL
gpt2	fp32 · yes	1e-3
pythia-160m	fp32 · yes	1e-3
Llama-3.2-1B	bf16 · no	1e-2
gemma-2-2b	fp32 · yes	1e-3
Qwen2.5-0.5B	bf16 · no	1e-2
gpt-oss-20b	bf16 · safe	1e-2

gemma-2 adds softcapping + sliding-window; gpt-oss-20b runs **safe_mode** (attention off, OOM guard) and **drops the first token** (attention-sink policy).

G0 · CALIBRATION SET

C-Set: EN / KO parallel

Four propositions in **translation-parallel** form (same claim, two languages), fixed as a reference set:

- C0 a small bird crosses a quiet garden
- C1 the system records several independent sensors
- C2 physics-guided ML gives interpretable representations
- C3 LLMs process information through hidden residual streams **(self-referential edge case)**

Protocol: **translation-parallel** first; a Korean-specific **structural_independent** phase is registered but **not written yet**. Eligible families (llama-3, llama-3.1, gemma-2, qwen-2.5) run the KO set; ineligible models **auto-skip with a logged reason**.

G0 · CONVENTIONS

Normalisation is data

Depth, position and density conventions are fixed once and reused by every later generation:

$$\text{rho} = l \,/\, \max(L,\,1) \quad \cdot \quad \text{rho} = l \,/\, \max(L-1,\,1)$$
$$\text{tau} = t \,/\, \max(T-1,\,1)$$

Residual index (0..L) and block-output index (0..L-1) use **different** normalisers; the first token is excluded for tau.

Length-matched strata: bins **16 / 32 / 64 / 128 / 256 / 512**, minimum 20 texts per bin — below that a bin is reported **INCONCLUSIVE**.

```
YAML is data — formulas are never eval'd; a wrong edit is
caught immediately.
```

G0 · PILOT

First run

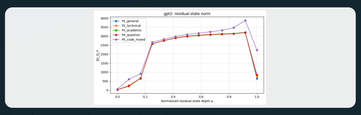


Fig. 1. G0 pilot output (Colab bundle, Qwen3 finalization v1.0.4).
Question that started it: are layer-wise norms and updates $\Delta\mathbf{h}_l = \mathbf{h}_{l+1} - \mathbf{h}_l$ finite and correctly shaped?

G0 · RIGOR

Rigor is a test suite

The measurement code itself is under test — **9 modules · 45 tests** guard the rig before any number is trusted:

- gates_yaml schema load · tolerance matches the precision table · **safe_mode disables attention capture** · rho variants agree
- normalize rho / tau endpoints · first token dropped · length-bin edges · entropy uniform = 1, degenerate = 0
- report bootstrap rejects n < 100 · **cluster-level**, not token-level · paired diff + true-null gate helper
- gpt-oss attention rows sum < 1 (sink) · router logits top-k · sliding-window mask beyond the window
- gemma softcapping path **exercised** · layer types alternate · real checkpoint test
- tiny / real gpt2 · llama · pythia tiny end-to-end runs, plus real-checkpoint tests

Eligibility is enforced in tests too: an **ineligible model is actually blocked**, never silently skipped.

N02 · WEEK 2

Three observables, frozen

gpt2 / **pythia-160m**, 12 blocks, instrumented block by block. The observable set is **frozen for traceability**:

1. length-adjusted update norm $\|h_{l+1} - h_l\|_F / \sqrt{T}$
2. input–update cosine $\cos(h_l,\ \Delta h_l)$
3. output/input norm ratio

Block outputs are measured **before** the final LayerNorm (logged separately).

N02 · RESULT

Cosine flips sign with depth

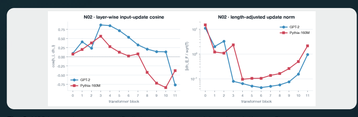


Fig. 2. N02 layer-wise summary (5 pilot prompts per model — descriptive, n=5).
Cosine rises, then **turns negative in the last blocks** (GPT-2 −0.769 at block 11; Pythia −0.840 at block 10).

N02 · RESULT

Attention vs MLP branch

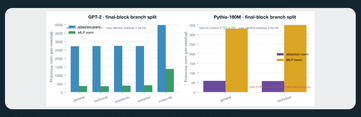


Fig. 3. Final-block branch split; identity residual ≤ 2.3e-5 (GPT-2).
Attention dominates GPT-2 (cosine 0.37–0.43) while Pythia is MLP-dominant (0.72–0.76) — one observable, two architectures.

N02 · PROTOCOL

Manifest before inference



Fig. 4. Candidate manifest v1.0.3 (preflight: 0 hard-check failures).

N02 · HONESTY

What the pilot is not

- The five pilot prompts are **excluded** from the main evaluation.
- The earlier synthetic chart was **not real model data**.
- N02's scientific hypothesis status stays **INCONCLUSIVE**; the baseline is frozen for traceability, not for claims.
- 24 code texts carry no domain word; 12 share one template.
- **frozen_for_confirmatory_analysis: false** — human review pending.

N03 · WEEK 3

Three branches, one active

N02 deferred three architecture families; N03 registers them and runs **M1 (MoE routing-conditional)**. M2 (MLA latent attention) and M3 (SSM / hybrid) stay **design-stage only** — no claim may use them yet.

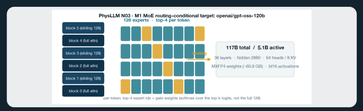


Fig. 5. Target: **openai/gpt-oss-120b** — MXFP4 weights, bf16 activations.

N03 · OBSERVABLES

Routing, not just residuals

Residual-stream measures are inherited from N02; N03 adds the branch split and the router:

- attn_share $\text{attn}^2/(\text{attn}^2+\text{moe}^2)$ of the pre-residual output
- routing top-4 expert ids + gate weights (softmax over the **top-k logits**, not the full 128)
- entropy routing entropy and H/ln k (k = 4)
- load per-text/layer expert load histogram

N03 · RESULT

Residual ratio falls, routing stays flat

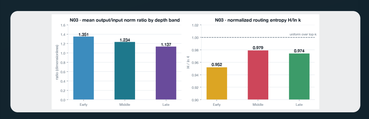


Fig. 6. Depth-band summary over 36 blocks; mean normalized entropy 0.968.
Output/input ratio **1.351 → 1.234 → 1.137** across early/middle/late blocks, while routing entropy stays near uniform (H/ln4 ≈ 0.95–0.98).

N03 · EVIDENCE

Gates, provenance, 80 GB wall

MXFP4 weights ≈ **60.8 GB**; 8K ctx ≈ 65.8 GB, 32K ≈ 68.6 GB — minimum **one 80 GB GPU** (A100 80GB / H100 80GB / MI300X); L4 24GB, T4 16GB, A100 40GB insufficient. Runtime: Triton ≥ 3.4, compute capability ≥ 7.5. Development uses gpt-oss-20b only for wiring.

Every capture passes the dataset gates (**G1, M1–M5, P1**) and ships manifest SHA-256, code SHA + git revision/dirty, tokenizer ID, activation dtype, seed 0, package versions. Resume is idempotent and refuses a foreign out-dir. **30 gate tests** pass, incl. an independent routing recomputation.

N03 · AUDIT

T0 and what is pending

The tokenizer audit recomputes token counts for every candidate: **gpt2** and **pythia-160m** clean (0 mismatches), but **gpt-oss leaves 3 long-tier texts out of band** (127–128 tokens vs a 129 floor) — not random: 11 of 33 long × question_dialogue texts sit within 2 tokens of the floor while other long styles clear it by 21+. Options A / B / B' / C are quantified in the decision memo — **choice pending**.

Done: scope, M1 capture script, gate tests, notebook simulator, real end-to-end capture on a tiny checkpoint, routing artifact for 400 texts, T0 audit. **Pending:** T0 choice · the paid 80 GB session · repo approval.

ACROSS GENERATIONS

What carries forward

One observable definition survives every generation:

$$\Delta h_l = h_{l+1} - h_l \quad \cdot \quad \|\Delta h_l\|_F/\sqrt{T} \quad \cdot \quad \cos(h_l,\ \Delta h_l)$$

G0 proved the rig is **lens-equivalent** across six checkpoints and two languages; N02 froze **what** to measure and **how** to sample; N03 reuses both and adds the router — same gates, same SHA-256 discipline, same habit of stating the limits next to the number.

Limits: exploratory descriptive statistics — no causal or topic-clustered inference · no independent validation against real router dispatch or an fp32 reference run · candidate manifest not frozen · max branch-identity residual 0.0187 (N03), 2.3e-5 (N02).

Artifacts: **physllm_g0** (G0 gate v1.0.2) · **physllmN02** · **PhysLLM_N03_Antigravity_Workspace**
All numbers are project measurements. Figures rendered from project artifacts; no project file was modified.